

ER diagram to Relational schema

```
CREATE TABLE TimePeriod (  
    FL_DATE DATE PRIMARY KEY,  
    YEAR SMALLINT NOT NULL,  
    MONTH VARCHAR(20) NOT NULL,  
    DAY_OF_MONTH TINYINT NOT NULL,  
    DAY_OF_WEEK VARCHAR(20) NOT NULL,  
    QUARTER VARCHAR(20) NOT NULL  
);
```

```
CREATE TABLE Airlines(  
    OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,  
    OP_UNIQUE_CARRIER VARCHAR(100) NOT NULL  
);
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CREATE TABLE Airports(  
    AIRPORT_ID INT PRIMARY KEY,  
    IATA_CODE VARCHAR(20) NOT NULL,  
    CITY_NAME VARCHAR(80) NOT NULL,  
    STATE_ABR VARCHAR(20) NOT NULL,  
    FOREIGN KEY (STATE_ABR) REFERENCES State(STATE_ABR)  
);
```

```
CREATE TABLE Flights (  
    FLIGHT_ID INT PRIMARY KEY,  
    OP_CARRIER_FL_NUM INT NOT NULL,  
    TAIL_NUM VARCHAR(20) NULL,  
    OP_CARRIER_AIRLINE_ID INT NOT NULL,
```

ROUTE_ID INT NOT NULL,
FL_DATE DATE NOT NULL,
CANCELLED TINYINT NOT NULL,
DIVERTED TINYINT NOT NULL,
FOREIGN KEY (OP_CARRIER_AIRLINE_ID) REFERENCES
Airlines(OP_CARRIER_AIRLINE_ID),
FOREIGN KEY (ROUTE_ID) REFERENCES Routes(ROUTE_ID),
FOREIGN KEY (FL_DATE) REFERENCES TimePeriod(FL_DATE)
);

CREATE TABLE FlightPerformance (
FLIGHT_ID INT PRIMARY KEY,
CRS_DEP_TIME TIME NOT NULL,
DEP_TIME TIME,
DEP_DELAY INT,
DEP_DELAY_NEW INT,
DEP_DEL15 TINYINT,
TAXI_OUT SMALLINT,
WHEELS_OFF TIME,
WHEELS_ON TIME,
TAXI_IN SMALLINT,
CRS_ARR_TIME TIME,
ARR_TIME TIME,
ARR_DELAY INT,
ARR_DELAY_NEW INT,

ARR_DEL15 TINYINT,
CRS_ELAPSED_TIME SMALLINT,
ACTUAL_ELAPSED_TIME SMALLINT,
AIR_TIME SMALLINT,
CANCELLATION_CODE CHAR(50),
DIV_AIRPORT_LANDINGS TINYINT,
DIV_REACHED_DEST TINYINT,
DIV_ACTUAL_ELAPSED_TIME TINYINT,
DIV_ARR_DELAY INT,
DIV_DISTANCE INT,
FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID),
FOREIGN KEY (CANCELLATION_CODE) REFERENCES
CancellationReason(CANCELLATION_CODE)
);

CREATE TABLE Routes (
ROUTE_ID INT PRIMARY KEY AUTO_INCREMENT,
ORIGIN_AIRPORT_ID INT NOT NULL,
DEST_AIRPORT_ID INT NOT NULL,
DISTANCE INT NOT NULL,
FOREIGN KEY (ORIGIN_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID),
FOREIGN KEY (DEST_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID),
);

CREATE TABLE FlightDelay (
DELAY_ID TINYINT NOT NULL,

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FLIGHT_ID INT NOT NULL,

MINUTES INT,

PRIMARY KEY (DELAY_ID, FLIGHT_ID),

FOREIGN KEY (DELAY_ID) REFERENCES DelayType(DELAY_ID),

FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID)

);

CREATE TABLE FlightDiversiion (

FLIGHT_ID INT NOT NULL,

DIVERSION_NUMBER TINYINT NOT NULL,

DIV_AIRPORT_ID INT NOT NULL,

DIV_TOTAL_GTIME SMALLINT NULL,

PRIMARY KEY (FLIGHT_ID, DIVERSION_NUMBER),

FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID),

FOREIGN KEY (DIV_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID)

);

```

```

CREATE TABLE State (

STATE_ABR CHAR(2) PRIMARY KEY,

STATE_NM VARCHAR(50) NOT NULL

);

```

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CREATE TABLE AirlineNames (

OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,

NAME VARCHAR(150) NOT NULL

);

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CREATE TABLE AirportNames (

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AIRPORT_ID INT PRIMARY KEY,  
NAME VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE DelayType (  
DELAY_ID TINYINT PRIMARY KEY,  
NAME VARCHAR(50) NOT NULL  
);
```

```
CREATE TABLE CancellationReason (  
CANCELLATION_CODE CHAR(1) PRIMARY KEY,  
DESCRIPTION VARCHAR(50) NOT NULL  
);
```